

PRABHAT THAKUR

Software Engineer | New Delhi, India

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SUMMARY

Full-stack engineer who owns systems end to end — schema design through AWS deployment. Now **engineering lead** for Amato Automotive India's digital products, where I rebuilt the corporate platform from scratch, shipped a multi-tenant franchise warranty platform and HR + learning platform, and now lead a group-wide MIS consolidating dealer-network data. Strong across **React/TypeScript, Python/FastAPI, Node.js/Express**, and **Java/Spring Boot**. Also the group's lead on **applied AI**, designing and running enablement programmes for sales, operations, design, and manufacturing teams across Amato Automotive.

EXPERIENCE

SOFTWARE ENGINEER | [Amato Automotive Pvt Ltd](#) | Dec 2024 – Present

- Serving as **Engineering Lead** for Amato Automotive's digital products — owning the full lifecycle from requirement gathering and architecture through deployment and maintenance, while mentoring the internal tech team across frontend, backend, and infrastructure.
- Leading development of a **group-wide Management Information System** unifying Sales, Leads, Finance, and OE Network data across a **1000+ dealer network** — closing the data-leakage and reporting gaps left by manual spreadsheet workflows (React/TypeScript, FastAPI/Python, PostgreSQL on a self-managed VPS with nginx and PM2).
- Engineered a **Google Sheets → PostgreSQL sync pipeline** that normalizes semi-structured spreadsheet data — inconsistent headers, mixed units, hand-typed dates — into a queryable schema with idempotent re-sync and full audit logging, reconciling **1,500+ dealer records across 5 OEMs** with verified zero data loss.
- Enforced **two-layer access control** (module-level and sheet-source-level) with **scoped JWT tokens across separate auth domains**, so a token from the public field-rep portal can never authenticate against the internal MIS; shipped an **OTP-based passwordless portal** for field reps with server-side record scoping and CSV/XLSX export.
- Rebuilt Amato AutomotiveIndia.com end to end** — frontend, backend, database architecture, and admin panel — replacing a legacy stack with a maintainable, performant platform.
- Managing and scaling **AWS infrastructure** (EC2, S3, IAM, CloudFront, Lambda) across all Amato Automotive digital platforms, covering deployments, access control, and uptime.
- Built and maintain the **Enterprise Franchise Management & Warranty Verification System**, a **Three.js 3D personalization configurator** for Mahindra accessories, and a responsive **Store Locator** with ratings and admin tooling.

SOFTWARE DEVELOPER TRAINEE | [RailWorld India Pvt Ltd](#) | Dec 2023 – Aug 2024

- Built backend services and **REST APIs** with Java, Spring Boot, and a microservices architecture, integrated with React.js frontends over MySQL.
- Delivered features, debugging, and performance work in an **Agile/Scrum** team — sprint planning, code review, and Git branching workflows.

AI ENABLEMENT & TRAINING

AI LITERACY PROGRAMME (Designed & Delivered) | [Amato Automotive Group](#) | Ongoing

- Designed and delivered a **company-wide AI literacy programme** for 500+ employees across Sales, Accounts, Design, and Operations — covering prompt engineering, deep research, and hands-on demos across **Claude, ChatGPT, Gemini, and NotebookLM**, with a custom prompt cheat sheet as post-session reference.
- Extended the programme **across Amato Automotive's manufacturing units**, running role-specific sessions across 4 units with production, planning, operations, and design teams on their own live data rather than generic examples
- Built **unit-level AI use cases** now in regular use — production planning and shift reporting, SOP and work-instruction drafting, operational data analysis in Excel, and AI-assisted concept imagery for product design.
- Packaged session output into **reusable prompt playbooks** and progressive workshop tracks, turning one-off training into a repeatable enablement programme across sites.

PROJECTS

Autoform Connect — Full-Stack HR Management System | React (Vite), TypeScript, Node.js, Express.js, Prisma ORM, MySQL, JWT, Vercel

Live: <https://connect.amatoautomotive.co.in/>

- Architected and shipped an **end-to-end HR platform** from scratch, designing a **38-table relational schema** spanning employees, attendance, leave, payroll, learning, documents, and performance.
- Secured the platform with **JWT over httpOnly cookies**, bcrypt hashing, OTP email password reset, four-tier **role-based access control** (HR / Manager / Leadership / Employee), and IP-aware rate limiting on login and sensitive endpoints.
- Built **Attendance, Leave, and Payroll** modules — clock-in/out with manager-approved corrections, multi-level approval chains (Employee → Manager → HR), salary-slip generation, and a **Tally CSV/Excel import pipeline**.
- Developed a full **Learning Management System** with multi-format modules (video, YouTube/Vimeo embeds, PDF, text), approval-based enrollment, and auto-issued certificates — gated by an **80% watch/read threshold** enforced via the YouTube IFrame API, Vimeo Player SDK, and Page Visibility API.
- Engineered a **proctored quiz engine** with a form-style builder, configurable pass marks, grade bands, and attempt limits, plus **tab-switch detection** (Page Visibility + blur) that auto-submits the attempt and alerts Manager/HR.
- Built an **E-Library** with in-app PDF viewing via blob-URL rendering for large files, and a **centralized notification service** with department/role/individual targeting behind a real-time unread-count bell.

Enterprise Franchise Management & Warranty Verification System | React (Vite), TypeScript, Tailwind, shadcn/ui, TanStack Query, Node.js, Express.js, MySQL, JWT

Live: <https://warranty2.autoformindia.co.in/>

- Built a **multi-tenant platform** serving Customers, Vendors, and Admins for high-value automotive warranty management (Seat Covers, PPF, EV) with JWT role-based access control.
- Engineered a **multi-stage fraud prevention pipeline** requiring verified installer-to-product-code linking before any warranty clears admin validation.
- Designed an **automated grievance redressal system** with department-routing algorithms that dispatch complaints to internal staff via secure tokenized email links.
- Audited and refactored **50+ complex SQL queries** — multi-table JOINS, dynamic filters — to parameterized bindings, eliminating SQL injection vectors and 500-level errors.
- Built a **franchise-to-distributor Order Management System** with webhook-based integration into existing internal systems, streamlining order placement and fulfillment tracking.

Mahindra & MG Accessory Personalisation Configurator | React, JavaScript, Express, Material UI, Canvas 2D, ONNX Runtime Web, WebAssembly, pdf-lib, jsPDF, html2canvas, Puppeteer

- Built a **React-based accessory personalisation configurator** for Mahindra & MG Hector accessories, supporting custom text, fonts, colours, artwork uploads, and front/rear seat placement previews.
- Developed a **Canvas 2D embroidery renderer** that converts user text into deterministic satin-stitch graphics using a custom stitch-sequencing algorithm for direction, spacing, and thread shading per glyph.
- Integrated browser-side **ONNX/WASM background removal** for uploaded artwork with lazy model loading, plus a **flood-fill (BFS-based) segmentation fallback**.
- Implemented **client-side PDF order generation** using pdf-lib and jsPDF with embedded fonts, product previews, customer details, and signature/dealership fields.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, Java, SQL

Frontend: React.js, Three.js, Tailwind CSS, shadcn/ui, TanStack Query, HTML5, CSS3

Backend: Node.js, Express.js, Prisma ORM, FastAPI, SQLAlchemy, Spring Boot, REST APIs, JWT, ONNX Runtime Web, WebAssembly

Databases & Cloud: PostgreSQL, MySQL, AWS (EC2, S3, IAM, CloudFront, Lambda), self-managed VPS (nginx, PM2),

AI: Prompt engineering Claude, OpenAI API, Google Gemini API, NotebookLM,

Tools & Libraries: Git, GitHub, VS Code, Postman, Jsoup, Apache POI, pdf-lib, jsPDF, Puppeteer

EDUCATION & CERTIFICATIONS

B.Tech – Computer Science & Engineering — Maharshi Dayanand University (GITAM, Jhajar, Haryana) | 2019–23 | 78.54%

CBSE Class XII — 70.2% (2019)

CBSE Class X — 86.54% (2017)

Full Stack Software Development — QSpiders / JSpiders, Delhi | Apr–Sep 2023